Automated University Course Scheduling

A Comparative Study of CSP, Local Search, and Q-Learning

N. Machichi U. Nkwinga B. Sogutlu G. Van den Eynde

Groupe 9 MA1 Ingénieur Civil, ULB

INFO-H410 - Techniques of Artificial Intelligence, 2025-2026

Outline

1. Problem Formulation
2. Approach A: CSP with Backtracking
3. Approach B: Local Search (Simulated Annealing)
4. Approach C: Q-Learning
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University Course Scheduling

Goal: assign each course a (room, timeslot) pair that satisfies all constraints.

Hard constraints (must all hold):

- **H1** No professor teaches two simultaneous courses
- **H2** No room hosts two simultaneous courses
- **H3** Room capacity $\geq$ course required capacity
- **H4** Timeslot $\in$ professor's availability

Soft constraints (maximised):

- **S1** Timeslot in professor's *preferred* slots
- **S2** No gap between courses in a day
- **S3** Daily load $\leq$ limit

Combined objective: $\mathcal{F}(\sigma) = -10 \cdot |\text{violations}| + \text{soft_score}(\sigma) \in (-\infty, 1]$

Search Space & Benchmark

Search space: $|\mathcal{R} \times \mathcal{T}|^{|\mathcal{C}|}$ — for 30 courses: $\approx 10^{70}$ assignments.

20 timeslots (5 days $\times$ 4 blocks of 2h), up to 8 rooms.

Instance	Courses	Profs	Rooms	Difficulty
instance_05	5	3	3	Toy
instance_15	15	6	5	Small
instance_30	30	10	8	Medium
instance_60	60	15	8	Medium-large
instance_100	100	20	8	Large
instance_125	125	25	8	Near CSP limit
instance_150	150	30	8	CSP failure zone

CSP: Backtracking + Heuristics

Each course is a **variable**; its domain is all valid (room, slot) pairs.

Variable ordering:

- **MRV** — pick the most constrained course first
- **Degree** — tie-break by number of conflicts with unassigned courses

Value ordering:

- **LCV** repurposed as soft-score optimiser: sort candidate values by *descending* soft score $\Rightarrow$ first solution found is already near-optimal

Constraint propagation:

- **Forward Checking** (FC): prune neighbours after each assignment, $O(d)$
- FC + MRV achieves **zero backtracks** up to 125 courses
- AC-3 tested but unnecessary overhead on our benchmark

CSP: Anytime Variant

Standard CSP: returns the first valid schedule found (already near-optimal).

Anytime CSP: continues beyond the first solution to improve the soft score until a time budget expires.

Key observation

Because LCV already steers the first solution toward high soft scores, the Anytime variant **never finds a meaningfully better second solution** within 30 s.

Timeout risk

If the budget expires before any solution is found (e.g. 30 s budget on instance_100 which needs 41 s), the Anytime variant returns **nothing** — worse than the standard solver.

Local Search: Hill Climbing & Simulated Annealing

Start: random complete schedule (constraints ignored).
Iterate: explore neighbours, accept or reject moves.

Two neighbourhood operators (50/50):

- **Move** — reassign one course to a random (slot, room)
- **Swap** — exchange assignments of two courses

Hill Climbing: only accept $\Delta\mathcal{F} > 0 \Rightarrow$ gets trapped in local optima.

Simulated Annealing: also accept worse moves with probability $e^{\Delta\mathcal{F}/T}$

- $T_0 = 10.0$, cooling rate $= 0.995$
- After 4 000 iterations: $T \approx 1.7 \times 10^{-9}$ (frozen)
- Returns the **best schedule seen** across all iterations

Q-Learning

MDP: place one course at a time; choose a (slot, room) action for the current course.

State: compact 5-feature vector (slot preference, professor/room conflicts, daily load, capacity).
Many distinct states share the same vector $\Rightarrow$ agent confuses situations.

Policy: ε-greedy, ε decays $1.0 \rightarrow 0.05$ over all episodes.

Q-value update (Bellman):

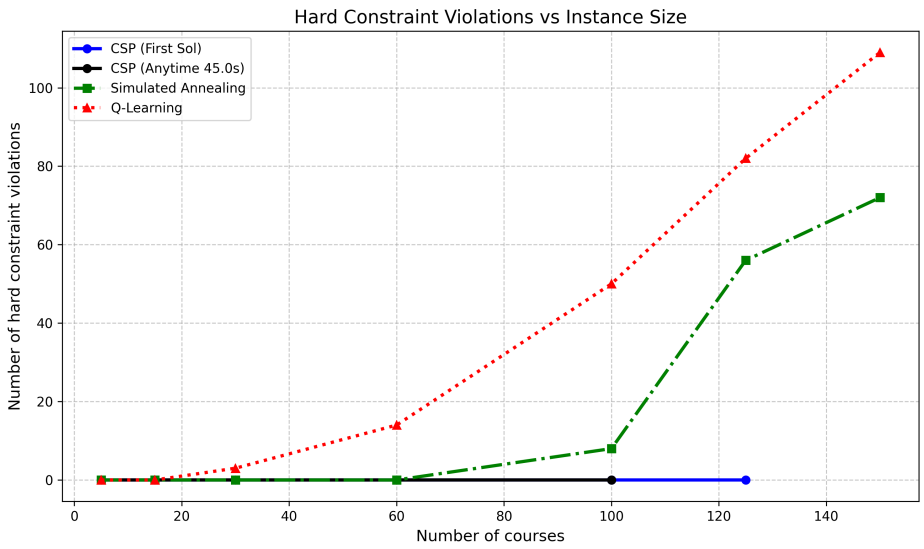
$$Q(s,a) \leftarrow Q(s,a) + \alpha\left[r + \gamma \max_{a'} Q(s',a') - Q(s,a)\right]$$

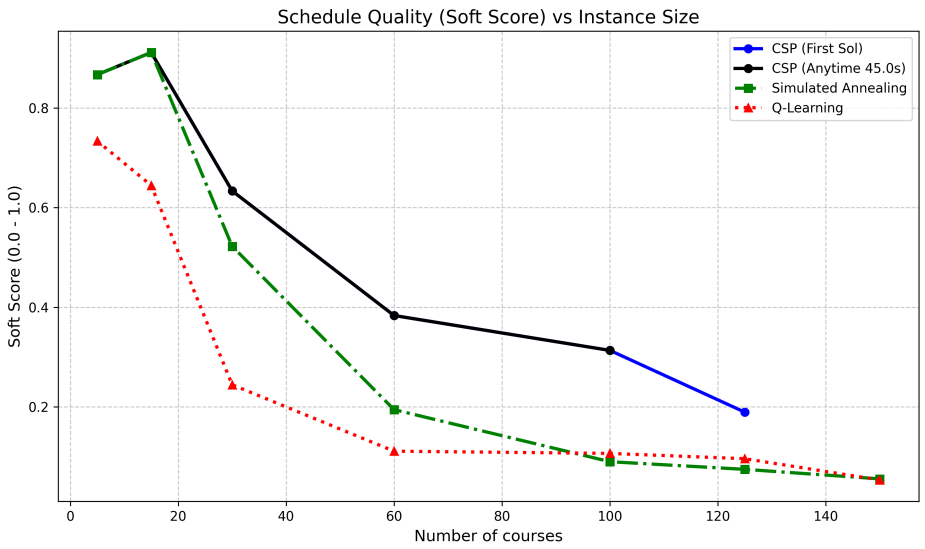
$\alpha = 0.1$, $\gamma = 0.9$, reward $r = \mathcal{F}(\text{current schedule})$.

Core limitation

Large instances get only 2 episodes — no opportunity to learn; decisions are essentially random.

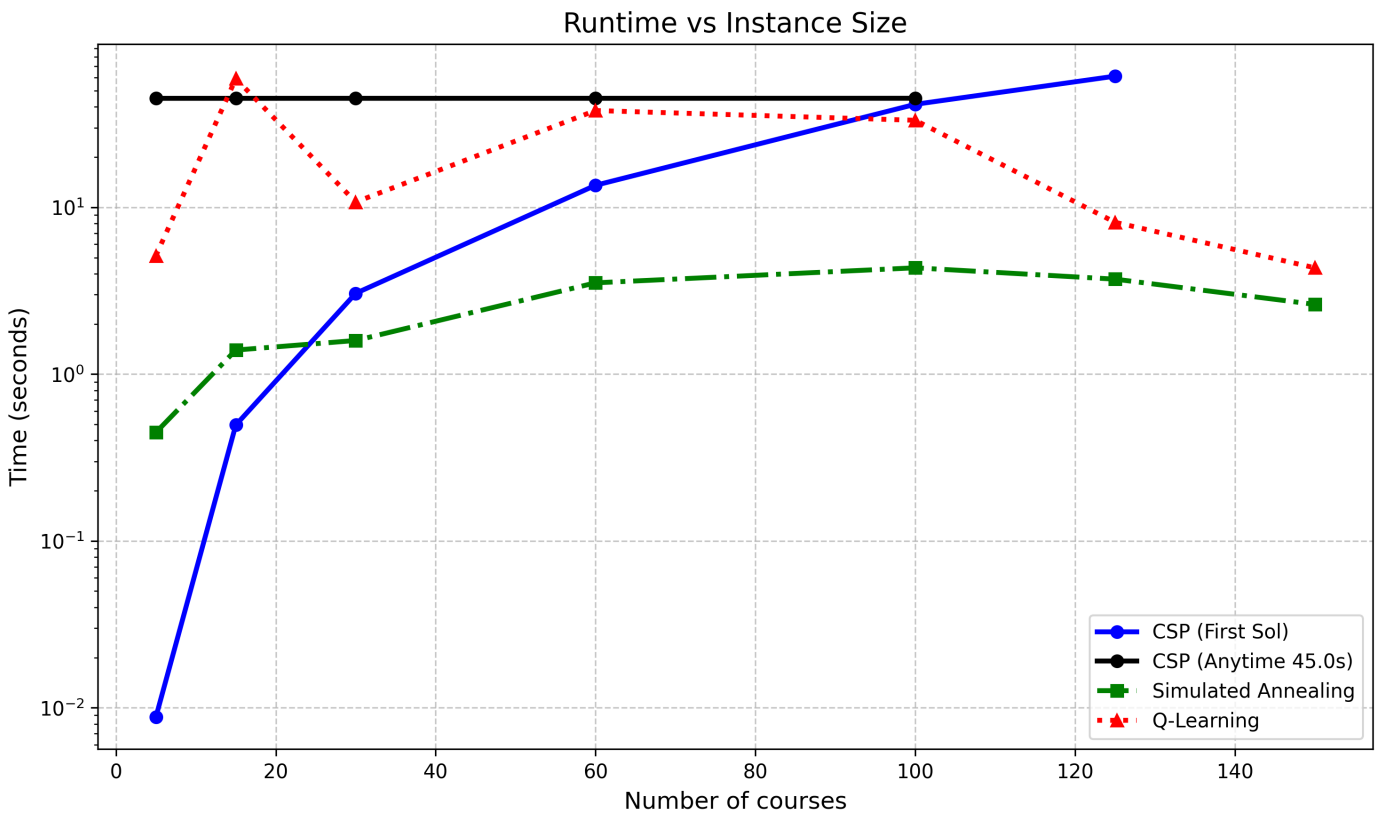
Results: Hard Violations & Solution Quality





- **CSP**: always 0 violations, highest soft score
- **SA**: 0 violations up to 60 courses, soft score drops from $n = 60$
- **Q-Learning**: violations explode from $n = 30$, soft score collapses

Results: Runtime



- **CSP**: 3 s at $n = 30 \rightarrow 61$ s at $n = 125$, hangs at $n = 150$
- **SA**: always < 5 s regardless of size
- **Q-Learning**: apparent speed-up on large instances is due to episode truncation (2 episodes), not efficiency

Qualitative Comparison

Criterion	CSP	SA	Q-Learning
Hard violations	**Always 0**	0 up to 60c	Grows from 30c
Soft score	**Highest**	Medium	Low
Scalability	Fails $>$ 125c	**All sizes**	Poor quality
Predictability	**Deterministic**	Stochastic	Highly variable
Hard guarantee	**Yes**	No	No

Take-away

- **CSP** when constraint satisfaction is non-negotiable ($n \leq 125$)
- **SA** as a practical all-round solver for any size
- **Q-Learning** not competitive in tabular form; motivates function approximation or hybrid approaches

Thank you for your attention

Code & data: `https://github.com/BurakSogutlu/Project-TAI-25-26-course_scheduler`